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Theme

Voice applications based on GenAI for
education: Empowering children with
ADHD through intelligent support

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Abstract

This project aims to develop an innovative digital solution designed to support children with Attention Deficit Hyperactivity Disorder (ADHD) throughout their educational and therapeutic journeys. Given the limitations of conventional care methods—particularly the lack of continuity between speech-language therapy sessions and the challenges of home-based monitoring for continuously busy parents—this proposed solution seeks to enhance therapeutic accompaniment by leveraging voice technologies and Generative Artificial Intelligence (GenAI). The system is centered around an intelligent voice assistant integrated into a smart speaker system, such as Alexa, which is seamlessly connected to a flexible web platform that systematically records comprehensive logs of the child’s learning and rehabilitation progress.

This voice-based interaction modality renders system utility highly accessible and natural, aligning perfectly with the cognitive and motor capacities of children. The child interacts organically with the assistant through quizzes, directed exercises, and educational games specifically engineered to stimulate attention and regulate impulsivity. Furthermore, the instructional and therapeutic content is automatically tailored to each child’s unique behavioral profile and learning trajectory via dynamic adaptation mechanisms powered by artificial intelligence.

Moreover, the platform features dedicated interfaces tailored for speech-language pathologists and parents, enabling them to create and manage comprehensive user profiles, design or dynamically generate targeted exercises utilizing GenAI, and monitor performance progression in real time through analytical dashboards and key performance indicators. This multi-stakeholder approach fosters close, synchronized, and highly effective collaboration among healthcare professionals, families, and the patient. By synthesizing voice assistants, smart hardware, generative AI, and robust analytical tracking tools, this solution establishes a complementary support ecosystem that bridges the gap in traditional clinical care, ultimately enhancing child engagement, guaranteeing continuity of care, and optimizing the overall efficacy of home-based therapeutic rehabilitation.

Résumé

Ce projet a pour ambition de développer une solution numérique innovante destinée à accompagner les enfants atteints du Trouble Déficit de l'Attention avec ou sans Hyperactivité (TDAH) tout au long de leur parcours éducatif et thérapeutique. Au regard des limites inhérentes aux modalités de prise en charge conventionnelles, en particulier l'absence de continuité entre les séances d'orthophonie et les difficultés de suivi à domicile pour des parents dont l'emploi du temps est constamment surchargé, la solution proposée vise à renforcer l'accompagnement thérapeutique en mobilisant les technologies vocales et l'Intelligence Artificielle Générative (IAGen). Le système s'articule autour d'un assistant vocal intelligent intégré à un dispositif d'enceinte connectée, à l'instar d'Alexa, lui-même relié à une plateforme web flexible qui consigne de manière systématique des journaux de bord exhaustifs retraçant la progression de l'enfant dans ses apprentissages et sa rééducation.

Cette modalité d'interaction vocale confère au système une accessibilité et une naturalité particulièrement élevées, s'accordant de façon optimale aux capacités cognitives et motrices des enfants. L'enfant interagit de manière spontanée avec l'assistant à travers des quiz, des exercices dirigés et des jeux éducatifs spécifiquement conçus pour stimuler l'attention et réguler l'impulsivité. En outre, les contenus pédagogiques et thérapeutiques sont automatiquement ajustés au profil comportemental singulier de chaque enfant ainsi qu'à sa trajectoire d'apprentissage individuelle, grâce à des mécanismes d'adaptation dynamique pilotés par l'intelligence artificielle.

La plateforme propose par ailleurs des interfaces dédiées aux orthophonistes et aux parents, leur permettant de créer et de gérer des profils utilisateurs complets, de concevoir ou de générer dynamiquement des exercices ciblés à l'aide de l'IAGen, ainsi que de suivre en temps réel l'évolution des performances au travers de tableaux de bord analytiques et d'indicateurs clés de performance. Cette approche multi-acteurs favorise une collaboration étroite, synchronisée et hautement efficace entre les professionnels de santé, les familles et le patient. En conjuguant assistants vocaux, dispositifs matériels connectés, intelligence artificielle générative et outils robustes de suivi analytique, cette solution constitue un écosystème de soutien complémentaire qui pallie les lacunes de la prise en charge

clinique traditionnelle, contribuant ainsi à renforcer l'engagement de l'enfant, à garantir la continuité des soins et à optimiser l'efficacité globale de la rééducation thérapeutique à domicile.

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Project Presentation

1.1 The Project Idea

The significant increase in the number of children with Attention Deficit Hyperactivity Disorder (TDAH) is a major public health challenge today. This disorder affects attention, impulse control, and executive functions, which leads to significant effects on a child's academic, social, and emotional development.

Although speech therapy sessions and specialized follow-ups are essential, they are often insufficient. This is due to their limited frequency, a lack of specialists in many areas that causes unequal access to services, a lack of continuous practice at home, and the daily difficulties families face in managing the child's condition.

Furthermore, turning to common digital alternatives like smartphones and tablets presents serious drawbacks. Extensive screen time from these devices can negatively impact children's developing brains, and their high retail prices make them a very expensive option compared to our affordable alternative.

In the face of these obstacles, providing an innovative solution to better support these children is necessary. This solution must ensure more regular monitoring, easier access, and smoother integration into their daily living environment without relying on harmful screen exposure.

The proposed solution relies on integrating an intelligent voice assistant designed to support children in their rehabilitation in an independent, interactive, and fun way. This project aims to modernize educational and therapeutic practices by taking advantage of advancements in artificial intelligence and voice technologies.

This solution makes it easy for the child to interact with the voice assistant, such as Alexa. With the help of generative artificial intelligence, the system creates customized quizzes and educational games that automatically adapt to each child's specific level, condition, and progress. Additionally, the platform allows real-time tracking of the child's progress, enabling parents and speech therapists to monitor them at any time and from

anywhere, providing complete and integrated support.

Thanks to this approach, therapy sessions become more flexible and accessible. This encourages continuous and personalized rehabilitation that effectively complements the traditional care provided by healthcare professionals.

1.2 Proposed Values

- **Time Efficiency and Continuity of Care:** Digital rehabilitation becomes readily accessible at home, facilitating consistent daily monitoring while eliminating the necessity for frequent clinical commutes and circumventing the scheduling constraints inherent to specialized centers.
- **Intelligent and Personalized Monitoring:** Leveraging artificial intelligence, the voice assistant dynamically generates tailored exercises and assessments based on the child's ongoing progress, effectively reinforcing areas of cognitive or linguistic weakness through targeted, purposeful activities.
- **Mitigation of Screen-Related Risks and Child Health Protection:** The solution relies entirely on auditory and vocal interaction, offering a health-conscious digital alternative that minimizes children's sedentary screen time (on smartphones and tablets). This approach safeguards the child's vision, enhances auditory comprehension and speech skills, and mitigates the adverse effects that excessive screen exposure has on focus and attention span.
- **Accessibility and Inclusivity:** Engineered to be simple and intuitive, the voice assistant ensures usability for children even in remote or structurally underserved regions that suffer from a deficit of adequate specialized care facilities.
- **Pedagogical Innovation:** The platform transforms therapeutic learning into an engaging, interactive, and motivating experience, thereby fostering the child's active participation, operational autonomy, and self-confidence.
- **Cost and Time Optimization:** Implementing this solution optimizes the tracking of the child's development while significantly lowering both the financial expenditures and the time commitments incurred by families.
- **Enhancing Speech-Language Pathology (Orthophonic) Monitoring:** The solution provides precise, structured analytics regarding the child's performance, progression, and specific challenges, thereby enabling clinical specialists to seamlessly adjust and optimize their therapeutic strategies.

1.3 The Team

1.3.1 Student 1 :Drizi Khaled

A Master 2 Computer Science student specializing in Information Systems Engineering (GSI) and a passionate Web Developer. Focused on cutting-edge technological solutions and artificial intelligence, he is dedicated to creating superior digital architectures that generate a positive impact on society.

Main rules and competencies:

- Responsible for the technical development of the project platform (architecture, implementation and integration of the various functionalities).
- Contribution to the integration of artificial intelligence mechanisms for monitoring and the adaptation of the exercises.

1.3.2 Student 2 : Adjou Maroua

A Master 2 Computer Science student specializing in Information Systems Engineering and Software (ISIL) and a passionate Web Developer. Driven by an interest in cutting-edge technological solutions and artificial intelligence, he focuses on the creation of superior, high-performance digital architectures.

Main rules and competencies :

- Formulated the go-to-market strategy, analyzed target user personas, and designed the competitive positioning for the platform.
- Spearheaded the creation of the project's pitch decks, visual assets, and comprehensive presentation materials.
- Able to work autonomously, resolve complex technical issues, and collaborate effectively in a multidisciplinary team.

1.3.3 Student 3 : Ghazi Manel :

A Master 2 Computer Science student specializing in Information Systems Engineering (GSI). She is highly interested in artificial intelligence, process automation, and Python development, and is currently expanding her technical expertise as an aspiring data analyst.

Main ruled and competencies :

- Acted as team lead, managing internal communications, task delegation, and cross-functional collaboration.

- Managed the financial modeling, market analysis, and budget structuring for the solution.
- Ability to work independently, solve complex technical problems, and collaborate effectively within a multidisciplinary team.

1.4 Project Objectives

The overarching goals of this project are structured around several core objectives:

1.4.1 Short-term objectives over the first 12 months :

- **Research and Development**
- **Consolidating Digital Presence and Expanding Market Reach:** Elevating the platform's online visibility and engagement through a comprehensive digital strategy. This encompasses targeted social media campaigns, search engine optimization (SEO), and the deployment of digital marketing tools designed to interface directly with primary stakeholder segments, specifically parents and healthcare practitioners.
- **Addressing Market Demands and Structural Gaps:** Responding to the escalating demand for digital intervention tools within the speech-language pathology (orthophonics) sector by delivering an accessible, customizable, and user-centric technological solution tailored to the distinct needs of children, parents, and practitioners alike.
- **Ensuring Continuity of Speech-Language Intervention:** Facilitating consistent, daily therapeutic practice via the voice assistant, thereby mitigating the risk of fragmentation or abrupt interruptions in the child's clinical care pathway.
- **Accelerating Market Share and Strategic Alliances:** Expanding the project's target audience through data-driven marketing frameworks and establishing strategic partnerships with speech-language pathologists, educational institutions, and specialized healthcare facilities.
- **Optimizing the Frequency and Efficacy of Therapeutic Sessions:** Establishing an efficient, continuous bridge between formal clinical consultations, thereby reinforcing behavioral and linguistic training to stimulate rapid, sustainable developmental progression in the child.

1.4.2 Long-term objectives between 2 and 5 years :

- **Geographical Expansion and Solution Deployment:** Expanding the platform's geographical footprint and penetrating new markets by scaling our network of speech-language pathologists (orthophonists) and establishing strategic collaborations with educational and medical institutions to accommodate the growing demand for digital, home-based therapeutic support.
- **Market Leadership and Positioning:** Attaining definitive market leadership within the sector of digital, home-based speech-language pathology and intervention services within the next five years.
- **Fostering Social Inclusivity:** Facilitating the academic integration and social inclusion of children experiencing language and communication disorders by providing consistent, data-driven, and highly personalized therapeutic tracking.
- **Sustainable Development and Technical Agility:** Implementing a continuous evaluation framework and iterative platform updates to ensure that the technological tools adapt to evolving user expectations and align with cutting-edge scientific paradigms.
- **Commitment to Quality Excellence:** Guaranteeing the delivery of an integrated digital therapeutic experience characterized by operational flexibility, robust efficacy, and precise customization tailored to the distinct needs of each child.
- **Consolidating Institutional Reputation and Credibility:** Establishing a distinguished brand identity as a reliable and authoritative benchmark in the market, anchored by high-quality service delivery and an unwavering commitment to professional standards and clinical ethics.
- **Sustaining Long-Term User Retention and Satisfaction:** Engineering a user-centric ecosystem that actively responds to the distinct needs of the primary stakeholder triad (child, parent, and practitioner) through continuous refinement of the platform's service design.
- **Maximizing Market Share and Portfolio Diversification:** Integrating innovative therapeutic solutions and sophisticated algorithms capable of addressing new developmental and behavioral disorders, thereby acquiring novel client segments while securing current user loyalty through operational excellence.

1.5 Project implementation schedule :

Nº	Steps	1	2	3	4	5	6	7	8	9	10
1	Feasibility study	✓									
2	Preliminary studies	✓	✓								
3	Prototyping		✓	✓							
4	Design and Modeling			✓	✓						
5	Backend Development (AI)				✓	✓					
6	Alexa Skill Development					✓	✓				
7	Testing and Integration							✓			
8	Activity launch								✓		
9	Marketing and Distribution									✓	
10	Presentation and Finalization										✓

Figure 1.1: Project implementation schedule

INNOVATIVE ASPECTS

2.1 Nature of Innovations :

Innovation refers to the successful introduction of new ideas, products, services, processes, or methods into the market or society. It aims to meet under-satisfied needs, resolve existing problems, or sustainably improve living conditions. The forms of innovation vary, ranging from gradual improvements (known as incremental innovations) to major transformations (or disruptive innovations) capable of reshaping entire sectors. Through their capacity to generate value, innovations play an essential role in economic growth, technological development, and social progress. They thus constitute a strategic lever for transforming our ways of life, our methods of working, and our interactions with the world around us.

The innovative points of this project are:

1. Child Experience & Personalized Learning

- Providing a daily, consistent, and customized training program that offers the flexibility to match each child's individual pace.
- Delivering a smart learning pathway that accurately diagnoses each child's specific challenges and reinforces their skills through targeted exercises and quizzes.

2. Professional Empowerment & Adaptability

- Equipping speech-language pathologists with an innovative digital platform for remote guidance, optimizing their clinical time and streamlining regular monitoring.
- Automating the generation of new therapeutic activities directly from materials (such as PDFs and text files) sent by the clinician, making the digital assistant

fully adaptable to actual case requirements.

3. Collaborative Monitoring & Seamless Adoption

- Generating detailed progress and activity reports accessible to both parents and professionals, ensuring rigorous educational and therapeutic oversight.
- Encouraging parents and professionals to smoothly adopt digital tools by naturally integrating them into the daily rehabilitation routine.

4. Security & Privacy

- Guaranteeing the absolute privacy and security of personal data through advanced encryption techniques, while fully complying with sensitive information protection standards.

2.2 Areas of Innovation :

2.2.1 New Features :

- **A New Experience for Children (The Core Focus):** Facilitating access to regular speech therapy training in a fun, interactive, and motivating format, thereby encouraging their engagement and accelerating their progress at home.
- **A New Experience for Parents (The Home Support):** Empowering parents to take a more active role in their child's rehabilitation, thanks to simplified access to exercises and progress reports, all via an intuitive interface that enables daily tracking without requiring technical expertise.
- **A New Experience for Speech therapists (The Professional Oversight):** Enabling professionals to monitor effectively from a distance, easily adjust exercises, and save time through automated tracking and activity-generation tools.

2.2.2 New pedagogical model :

- **Utilizing Technology in Speech Rehabilitation:** Effectively and seamlessly connecting children, parents, and speech-language pathologists through a specialized platform, optimizing session management, personalized progress tracking, and the coordination of therapeutic interventions.
- **Automated Generation of Personalized Content:** Automatically generating exercises and games tailored to the specific level and unique needs of each child, powered by artificial intelligence and Retrieval-Augmented Generation (RAG) technology.

- **Advanced Personalization:** Adapting not only the difficulty level of the exercises but also their format, duration, and frequency based on each child’s cognitive profile and learning pace, thereby delivering a fully custom-tailored experience.
- **Intelligent Notifications and Automated Reminders:** Sending personalized reminders and words of encouragement to maintain session consistency and boost the child’s motivation.
- **Advanced Tracking and Performance Analysis:** Providing detailed dashboards that showcase progress, highlight difficulties, and offer adaptation recommendations, allowing both speech pathologists and parents to effectively monitor the child’s evolution and make well-informed decisions.

2.2.3 Future vision:

- **Technological Evolution:** Integrating advanced artificial intelligence features, including speech recognition, emotional analysis, and predictive exercise adaptation, in order to enhance the personalization and effectiveness of sessions.
- **Launch of the Mobile Version:** Developing a comprehensive application for Android and iOS, offering flexible, interactive, and continuous access to exercises and educational activities.
- **Strategic Partnerships:** Collaborating with speech-language pathologists and specialized centers to enrich content, validate exercises, and ensure the clinical quality of the platform.
- **Sustainability and Continuous Improvement:** Securely collecting anonymized data to analyze the effectiveness of interventions, improve AI models, and adapt the platform to the evolving needs of users.
- **Extension of Services:** Future onboarding and management of other developmental and learning disorders, such as learning difficulties, dyslexia, and social communication disorders.
- **Broadening the Scope:** Progressive deployment across the entire national territory, with the ambition of international expansion tailored to the linguistic and cultural contexts of the target countries.

STRATEGIC MARKET ANALYSIS

3.1 The market segment:

3.1.1 The Potential Market:

Our project operates within a highly promising and rapidly expanding market segment, positioned at the intersection of three strategic and vital areas: digital health, remote speech therapy rehabilitation, and applied educational technologies.

Attention Deficit Hyperactivity Disorder (ADHD) is classified as one of the most common neurodevelopmental disorders among both children and adults. According to baseline indicators from the World Health Organization (WHO), the global prevalence of this disorder is estimated to be between 5% and 7% in children. Projecting these rates onto Algeria’s current demographic landscape—where the population has surpassed 47 million—we are looking at a target demographic exceeding 600,000 potential cases among children. Furthermore, approximately 50% to 60% of those diagnosed continue to experience symptoms into adulthood, representing a prevalence rate of 2.5% to 4.4% among the adult population.

Despite these figures, a vast segment of this population lacks structured care or specialized monitoring. This creates a clear, sustainable, and annually renewing target market that carries deep social and human significance. This reality provides our digital solution with robust, long-term commercial and investment potential.

In parallel, Algeria is experiencing a rapid digital transformation in the healthcare and education sectors. This boom is driven by the widespread expansion of high-speed mobile internet, increasing household reliance on smart devices, and the emergence of a new generation of tech-savvy parents who are entirely open to utilizing digital solutions to support their children’s development.

Conversely, currently available local solutions face a clear deficit in meeting the precise requirements of children with ADHD. It is exceptionally rare to find tools that intelligently and contextually integrate advanced features—such as AI-driven content generation, voice

interaction, and personalized speech therapy monitoring—while remaining fully adapted to the linguistic and cultural identity of Algerian society.

Together, these elements confirm the exceptional strategic relevance of our project within an investment environment characterized by clearly defined, genuine needs, fragmented existing solutions, and vastly open spaces for innovation.

3.1.2 The target market:

The Target Market: The target market represents a more refined and directly addressed segment, specifically tailored to our digital solution. For our intelligent voice assistant dedicated to home-based speech rehabilitation, the target market includes:

- **Parents of children with ADHD:** Who face practical challenges in ensuring consistent rehabilitation at home due to a lack of resources, time constraints, or limited access to specialized professionals, particularly in rural or remote areas.
- **Speech-Language Pathologists and Healthcare Professionals:** Who wish to extend their care remotely and acquire reliable tracking data to adapt their interventions effectively.
- **Educational Institutions and Specialized Centers:** Catering to children with special educational needs that are open to integrating supportive digital solutions into their programs.
- **Social Structures and Community Associations:** Active in the fields of social inclusion, disability support, or family assistance, seeking adaptable tools tailored to the local context.

3.2 Competitor Analysis :

3.2.1 Direct Competitors :

Our concept has a significant competitive advantage because it is positioned as a pioneering invention on the Algerian market, where no direct counterpart is currently listed. On a global scale, there are three obvious applications:

- **Abjd:** An innovative local project that represents a university graduation thesis idea supported only by a prototype, without having officially obtained the "innovative project" label yet. This project focuses on the concept of using voice technologies or intelligent assistants to teach foreign languages to children; however, it lacks the integrated infrastructure, continuous technical support, and advanced customization mechanisms.

- **Ari:** An application that uses artificial intelligence to monitor attentional functions in children with ADHD. It offers personalized exercises and activities but does not allow advanced voice interaction or integrated tracking by healthcare professionals.
- **Summit:** An educational platform for children with attention disorders, offering cognitive exercise programs and performance tracking. However, content adaptation remains limited, and voice interaction is not fully exploited.
- **Dopamind:** A solution focused on optimizing executive and behavioral functions through interactive games and exercises. Although efficient on a cognitive level, it does not provide integrated speech therapy tracking or full personalization of content.
- **Quizlet:** A digital study platform utilizing flashcards and automated quizzes; however, it lacks native audio reading capabilities.

3.2.2 Indirect Competitors :

- **Conventional, Non-Digital Methodologies:** This category encompasses families and speech-language pathologists who rely on traditional resources—such as printed materials, books, and in-person therapeutic interventions. While foundational, these approaches offer constrained oversight within the home environment, lacking both automated interactive features and empirical, data-driven analytics to systematically monitor the child’s developmental progress.
- **Generic Speech-Language Therapy Applications:** This category includes broad-spectrum digital tools, such as Speech Blubs or OrthoPicto, designed to provide general language exercises and gamified learning activities. However, these platforms lack the specialized framework required to support children with Attention Deficit Hyperactivity Disorder (ADHD) and fail to offer a comprehensive, tailored therapeutic regimen.

The strengths and weaknesses of competitors:

Competitors' Strengths	Competitors' Weaknesses
Experience and maturity in the market.	Limited or non-existent performance analysis.
Solidly established reputation.	Low personalization; standardized exercises for everyone.
Local trust and credibility.	Incomplete progress tracking for parents and professionals.
Active user base and actionable user feedback.	Simple, sometimes absent reports.
	Difficulty tracking long-term progress.
	Excessive screen time within the applications.

Figure 3.1: Strengths and weaknesses of our competitors

3.3 SWOT Analysis :

The SWOT analysis is a standard strategic framework widely deployed by enterprises and project developers to evaluate market positioning, growth potential, and competitive advantage. It is anchored in a structured methodology designed to identify and analyze internal strengths and weaknesses alongside external opportunities and threats that could critically influence the project's ultimate success.

- **Strengths:** Internal, positive attributes that confer a distinct competitive edge upon the enterprise or project. This includes core strategic resources, specialized institutional competencies, proprietary or innovative technologies, domain expertise, and operational efficiency.
- **Weaknesses:** Internal limitations and structural constraints capable of impeding overall performance or complicating project execution. These typically encompass resource deficits, organizational or operational gaps, restricted industry experience, or over-reliance on specific technological partners.
- **Opportunities:** Favorable external dynamics that the enterprise can strategically exploit to accelerate its growth and development trajectory. This dimension encompasses evolving market trends, technological advancements, legislative or regulatory shifts, and emerging consumer demands.
- **Threats:** Adverse external variables and environmental constraints that pose risks to project viability if not preemptively mitigated. These may manifest as intensifying market competition, regulatory obstacles, macroeconomic instability, or low adoption rates among the target demographic.

The SWOT model for our company:

- **Comprehensive User Research and Needs Analysis:** Conducting methodical empirical investigations into the behaviors, requirements, and workflows of final users (children, parents, and healthcare practitioners) to precisely calibrate and align the platform’s functional and technical specifications with their objective clinical and domestic needs.
- **Strategic Marketing Formulation:** Constructing a coherent and integrated marketing framework defined by explicit, quantifiable, and standardized performance indicators (KPIs) that are rigorously aligned with the structural dynamics and expectations of the target market.
- **Brand Identity and Value Proposition Consolidation:** Establishing a distinct and robust institutional brand identity by systematically highlighting the tripartite dimensions—humanitarian, social, and technological—of the digital solution to foster trust among parents, healthcare professionals, and relevant institutional bodies.
- **Systematic Competitive Intelligence:** Executing continuous and structured competitive assessments to identify market gaps, operational limitations of existing solutions, and strategic levers for differentiation, thereby fortifying the enterprise’s sustainable competitive advantage.
- **Targeted User Acquisition and Growth Initiatives:** Deploying precise, data-driven promotional actions and strategic marketing campaigns optimized to enhance platform attractiveness, expand market penetration, and capture new segments of active users.

3.3.1 Purpose of the SWOT Analysis:

- **Comprehensive Strategic Diagnosis:** Execution of a holistic strategic assessment to identify internal determinants that either catalyze or impede project performance, while concurrently evaluating external environmental dynamics capable of altering its evolutionary trajectory.
- **Identification and Valorization of Distinctive Competitive Advantages:** Highlighting the project’s unique competitive positioning by leveraging core strategic resources, technological infrastructure, and organizational competencies, measured objectively against the market positioning of key competitors.

- **Preemptive Mitigation of Structural and Cyclical Risks:** Anticipating latent market, regulatory, competitive, and behavioral risks to design and implement robust, proactive risk-mitigation frameworks.
- **Strategic Capitalization on Growth Opportunities:** Identifying and exploiting favorable external conditions—including macroeconomic trends, unaddressed market demands, and institutional support mechanisms—to guide core strategic choices and fortify long-term organizational viability.

3.3.2 SWOT matrix:



Figure 3.2: SWOT

3.4 The Marketing Strategy:

A marketing strategy is a comprehensive plan that outlines how a business will promote its products or services to its target audience, including goals, tactics, and metrics to achieve specific objectives like increased brand visibility and customer engagement. It serves as a blueprint for businesses to allocate resources effectively and respond to market changes.

3.4.1 The Marketing Strategy for our company :

3.4.2 The Marketing Mix (7P):

The marketing mix encompasses the comprehensive set of strategic levers mobilized by an enterprise to effectively position its value proposition within the market. It serves as a structured framework to align decisions regarding the product, its pricing architecture, distribution channels, and promotional strategies to ensure commercial viability and market adoption.

1. **Product:** The digital solution is configured as a website dedicated to the speech-language rehabilitation of children diagnosed with Attention Deficit Hyperactivity Disorder (ADHD). It integrates an intelligent voice assistant capable of interacting dynamically with the child, generating profile-adaptive exercises, and tracking cognitive progress. Concurrently, dedicated dashboards allow parents and speech-language pathologists to customize therapeutic content and conduct remote tele-monitoring.
2. **Price:** A freemium model featuring a defined initial complimentary access period will be deployed to lower entry barriers and catalyze user adoption. This is followed by a strategically calculated tier-pricing structure designed to maintain socio-economic accessibility while guaranteeing the long-term financial sustainability and operational continuity of the service.
3. **Place:** The platform is cloud-based and accessible via standard web browsers across multiple hardware configurations, including personal computers, tablets, and smartphones. Market penetration will be progressively expanded through B2B institutional partnerships with clinical speech therapists and specialized educational institutions to secure targeted visibility among end-users.
4. **Promotion:** This dimension relies on an omni-channel marketing strategy combining digital communication, authoritative psychoeducational content creation, and institutional networking. An active social media presence will drive community sensitization regarding ADHD, supported by targeted, data-driven advertising campaigns (Google Ads, Meta Ads). Furthermore, collaborative endorsements from clinical professionals, educational bodies, and healthcare associations will reinforce the solution's academic legitimacy and organic adoption.
5. **People:** This represents the human capital and key stakeholders contributing to service delivery and value co-creation. It specifically encompasses software engineers, UI/UX designers, certified speech-language pathologists, and specialized consultants whose collective expertise ensures the technical integrity and clinical efficacy of the platform.

6. **Process:** The solution optimizes the user journey through a seamless, intuitive operational workflow. The child is autonomously guided via synchronized vocal cues through personalized therapeutic activities, while parents and clinicians interface with a minimalist, ergonomic dashboard optimized for rapid session tracking and real-time adjustments.
7. **Physical Evidence:** This constitutes the tangible and technical ecosystem validating service quality. It comprises the underlying computational infrastructure, robust integrated development environments (IDEs), cloud hosting servers, and proprietary voice-processing and artificial intelligence (AI) integration tools that collectively guarantee system performance and data security.

PRODUCTION STRATEGY AND ORGANIZATION PLAN

4.1 The Process of Obtaining the Service

Access to the service is simple, seamless, and secure through a dual-interface ecosystem. While children engage in educational sessions exclusively through an interactive voice assistant, parents and speech therapists can access a dedicated web platform to monitor progress records. The entire workflow is designed to ensure a quick learning curve for all user profiles while accommodating the specific cognitive needs associated with attention disorders.

4.1.1 Concerning the child:

- **Account Creation for Parents and Speech therapists:** accounts are created exclusively by parents or speech therapist. These accounts integrate critical baseline data ,such as the child’s age, clinical diagnosis, baseline difficulty level, and historical performance ,to tailor the learning trajectory. No individual child accounts are generated,just adding children related to each parent (only the voice assistant is used for the child).
- **Voice Skill Activation:** The interaction session is initiated directly via a skill on the voice assistant. The user interface is intentionally minimalist to foster cognitive autonomy, optimize comprehension, and ensure intuitive usability for the child.
- **Child Response and Skill Termination:** The child engages with exercises, vocal quizzes, and guided attention tasks. Upon the completion of these interactive activities, the voice skill session terminates automatically.
- **Real-Time Progress Acquisition and Dashboard Analytics:** Throughout

the session, the system automatically quantifies the child’s behavioral responses, linguistic progress, and specific cognitive friction points in real time. Rather than maintaining a child-facing profile, these analytical insights are populated directly onto a centralized dashboard, enabling parents and clinicians to monitor the child’s developmental trajectory continuously.

4.1.2 Concerning Speech therapists and health professionals:

- **Professional Account Creation and Management:** Each speech therapists or healthcare practitioner is provisioned with a secure workspace. Within this environment, the clinician initiates and provisions accounts for the parents under their care, customizes rehabilitation pathways, and accesses macro-level performance analytics tools.
- **Parent-Managed Child Data Integration:** Upon account activation, the parent can input and modify their children’s clinical and demographic information directly within their own user profile. This completely eliminates the technical requirement for generating separate, standalone accounts or profiles for the children.
- **Performance Monitoring and Analytics:** Clinicians can monitor each child’s quantitative progress, specific cognitive friction points, and behavioral markers in real time. This is facilitated through detailed statistical metrics and synthesized graphical visualizations, optimizing both diagnostic evaluation and intervention planning.
- **Customized Exercise Calibration:** Data captured during active sessions enables the speech therapists to dynamically adapt exercises, vocal quizzes, and cognitive activities based on the unique needs of each child (as documented within the parent’s profile), thereby maximizing the efficacy of the therapeutic regimen.
- **Parent-Clinician Communication and Coordination:** The platform incorporates a secure communication channel designed to facilitate dialogue between the speech therapist and parents. This channel supports the dissemination of clinical recommendations and observational notes, ensuring a coherent and structured continuum of care at home.
- **Artificial Intelligence-Assisted Diagnostics:** The integrated AI engine assists speech therapists by automatically analyzing performance metrics, identifying specific linguistic or attention deficits, and algorithmically proposing tactical adjustments to guarantee highly precise, individualized patient monitoring.

4.1.3 Concerning the parents :

- **Parent Account Management and Child Data Integration:** The parent utilizes a secure workspace (previously provisioned by the speech therapist) that allows them to input and manage their children’s data directly from within their own user profile, configure account parameters, and access various monitoring features. The interface is intentionally intuitive, engineered to facilitate simple and rapid navigation.
- **Rehabilitation Pathway Configuration:** In coordination with the clinician, the parent can modulate specific training and rehabilitation parameters—such as target objectives, difficulty levels, session frequency, or scheduled usage windows—to seamlessly adapt the intervention to the family dynamic and the child’s specific developmental needs.
- **Access to the Monitoring Dashboard:** A synthesized analytics dashboard aggregates data on progress achieved, targeted competencies, specific cognitive or linguistic friction points detected by the AI, average attention span, and longitudinal performance trends over weeks or months. This monitoring system empowers parents to better comprehend their child’s needs and systematically support their progression at home.
- **Intelligent Notifications and Personalized Recommendations:** The parent receives automated notifications regarding milestones achieved, potential indicators of regression, recommended supplementary exercises, and session reminders. Concurrently, the integrated AI generates actionable insights to optimize the child’s environmental conditions at home to enhance cognitive focus and attention.
- **Clinician-Parent Communication Continuum:** The platform features a secure communication channel dedicated to interacting with the Speech-Language Pathologist. This allows parents to share observational notes, ask questions, and review professional clinical feedback, thereby reinforcing the therapeutic coordination between the family unit and the specialist.

4.2 Procurement:

Procurement refers to the comprehensive assembly of technical, human, and organizational resources essential for the seamless operation of a service. Within the context of a digital solution, it specifically encompasses the software, hardware, and infrastructural components required for both its development and long-term sustainability. For this project, the procurement framework is structured around the following core pillars:

- **Development Infrastructure and Collaborative Environment:** In the initial phase, development leverages the physical and technical infrastructure provided by university research laboratories. This strategic approach minimizes fixed overhead costs while establishing a highly collaborative environment optimal for an efficient project launch.
- **Software Tools and Licensing Frameworks:** To optimize development costs, the team utilizes open-source software and educational licensing frameworks (including Visual Studio Code ...). The Large Language Model (LLM) is integrated via an Application Programming Interface (API) or a cloud-based license, while the Alexa voice assistant is engineered using the Amazon Developer Console to ensure natural, seamless, and continuous verbal interaction.
- **Cloud Services and Web Hosting:** The web application, database architecture, and backend APIs are hosted on reliable cloud infrastructures . This ensures robust security protocols, high availability, and horizontal scalability aligned with the evolving requirements of the project.
- **Academic Partnerships and Institutional Support:** The project is officially endorsed by the partner university and its technological incubator. This partnership grants the team vital access to educational resources, hardware, scientific mentorship, and specialized training programs, thereby accelerating the development and maturation of the digital solution.

4.3 Workforce:

The project outlines the creation of 8 direct jobs during its first fiscal year (N+1). Operationalizing the service will mobilize specialized technical capabilities—specifically AI/Full-stack development and network engineering. To ensure comprehensive organizational management, the core technical team will be supported by essential cross-functional roles in marketing, pedagogy/language expertise (speech therapy), accounting, and administration.

The initial workforce will begin at 8 employees and follow a structured annual progression aligned with the scaling dynamics of the business. Consequently, the company's headcount will expand to 12 employees in the second year (N+2), steadily growing to reach a total of 20 employees by the fifth year (N+5).

A detailed forward-looking projection of the workforce distribution, broken down by specific organizational roles and fiscal years, is illustrated below:

Role	N+1	N+2	N+3	N+4	N+5
Founder	2	2	2	2	2
AI / Fullstack Developer	2	3	4	5	5
Developer	1	2	3	3	4
Network Engineer	0	1	1	1	1
Marketing Manager	1	1	1	1	1
Pedagogue / Language Expert	2	2	2	2	3
Accountant	0	0	1	1	1
Secretariat / Administration	0	1	1	1	1
HRM (HR Manager)	0	0	0	1	1
Total	8	12	15	17	20

Figure 4.1: Personnel Headcount Estimation over Five Years

4.4 Key partners :

1. Speech Therapists and Healthcare Professionals:

- Role: To seamlessly integrate the solution into clinical practices, collaborate on the formulation and validation of therapeutic content, and provide ongoing field feedback to drive continuous optimization of the tool based on practical clinical requirements.

2. Educational Institutions and Specialized Schools:

- Role: To cooperate in field-testing and piloting the solution, help identify the specific cognitive and behavioral needs of children, and facilitate the adoption of the tool within specialized learning and rehabilitation curricula.

3. Financial Partners and Incubators:

- Role: To provide initial seed funding, guide project execution through its foundational development phases, and accelerate strategic networking with key actors across the educational, medical, and technological ecosystems.

EXPERIMENTAL PROTOTYPE

5.1 Introduction :

The primary objective of this chapter is to detail the technical realization of the Atheeria , ADHD Child platformv, an interactive, multi-dimensional educational and therapeutic solution tailored for children with Attention Deficit Hyperactivity Disorder (ADHD). Moving beyond the previously established theoretical frameworks, user needs analyses, and architectural designs, this phase serves to validate our technical methodologies. We begin by outlining the hardware and software environments that facilitated development. Subsequently, we provide an in-depth examination of the deployed system architecture. Finally, we present the specialized user interfaces designed for children, parents, and physicians, supplemented by rigorous technical analyses, and operational screenshots of the functioning prototype.

5.2 Development Environment, Tools and Technologies:

We have chosen a range of cutting-edge and reliable technologies to satisfy the performance, modularity, and security needs.

5.2.1 Hardware Environment and Publishers:

Development was carried out on computers running Windows 10/11 and Linux, using Cursor ,an editor renowned for its rich extension ecosystem and native terminal integration.

5.2.2 Collaborative Tools and Version Control

In order to ensure the code's continuity and the security of its implementations, we have relied on:

- **Git:** versioning tool for tracing code changes source
- **Github :** Cloud platform hosting our main repository, making it easy to deploy Continuous

5.2.3 Client-side technologies (Front-end):

Deployed on Vercel , benefiting from edge caching, automatic SSL, and instant roll-backs. The SPA architecture ensures fast page transitions without full reloads.

5.2.4 Server-side (back-end) and cloud technologies:

The backend architecture relies on a decoupled, service-oriented design to handle distinct workloads:

- **LLM API (Groq):** Powers MCQ generation from uploaded PDFs. Groq was selected for its high-speed inference, enabling real-time quiz creation without noticeable latency for end users.
- **PostgreSQL (Render managed):** Central relational database storing user accounts, quiz history, progress tracking, and session data. Render's managed service handles backups, scaling, and maintenance automatically.
- **Authentication:** Built from scratch using bcrypt for password hashing. No third-party provider (Auth0, Firebase) , keeping dependencies minimal and costs low.
- **Alexa platform:** Developer Console (free tier) used for skill publishing, testing, and certification before public release.

5.3 Present and Implementation of the users Interface

5.3.1 Platform home interface

These images show the homepage of our platform, which is clear and well-organized, creating a positive and appealing first impression for users.

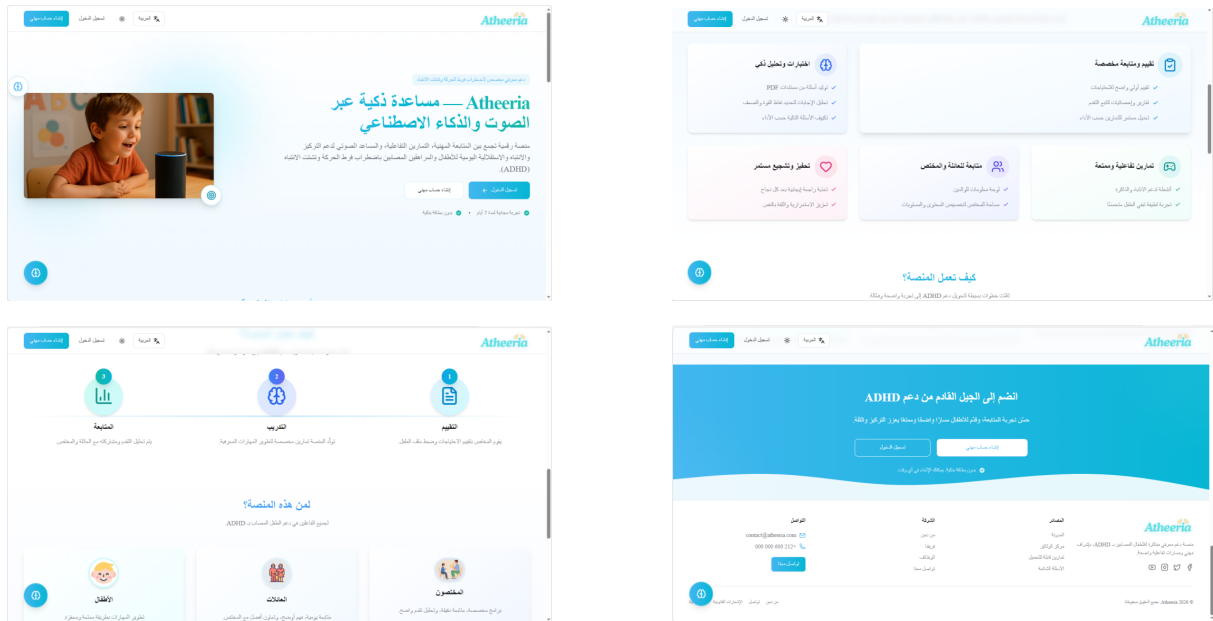


Figure 5.1: Platform home interface

5.3.2 Speech Therapist Dashboard

This dashboard provides a comprehensive overview of platform activity, including key metrics (children enrolled, progress rates, retention rates) and trend charts. This enables close monitoring by speech therapists.

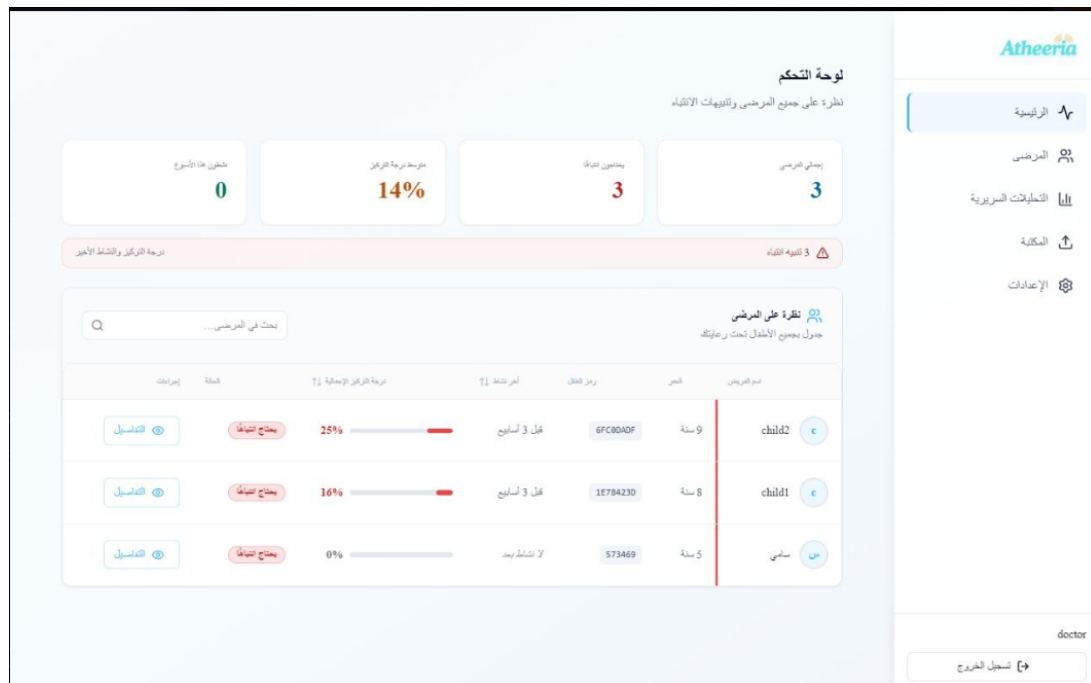


Figure 5.2: Dashboard



Figure 5.3: Dashboard

5.3.3 Interface to add a new child :

Adding a new child by the doctor on the website: A privileged interface that allows a verified doctor to register a new child into the system, including personal details, and guardian information, directly through the web portal.

Figure 5.4: Interface for adding a new child

5.3.4 Dashboard to monitor the child's progress by the specialist :

The specialist dashboard enables real-time tracking of a child's developmental progress. It visualizes performance graphs, attention metrics, quiz results, and exercise completion rates, helping specialists evaluate therapy effectiveness, identify intervention needs, and personalize treatment strategies.

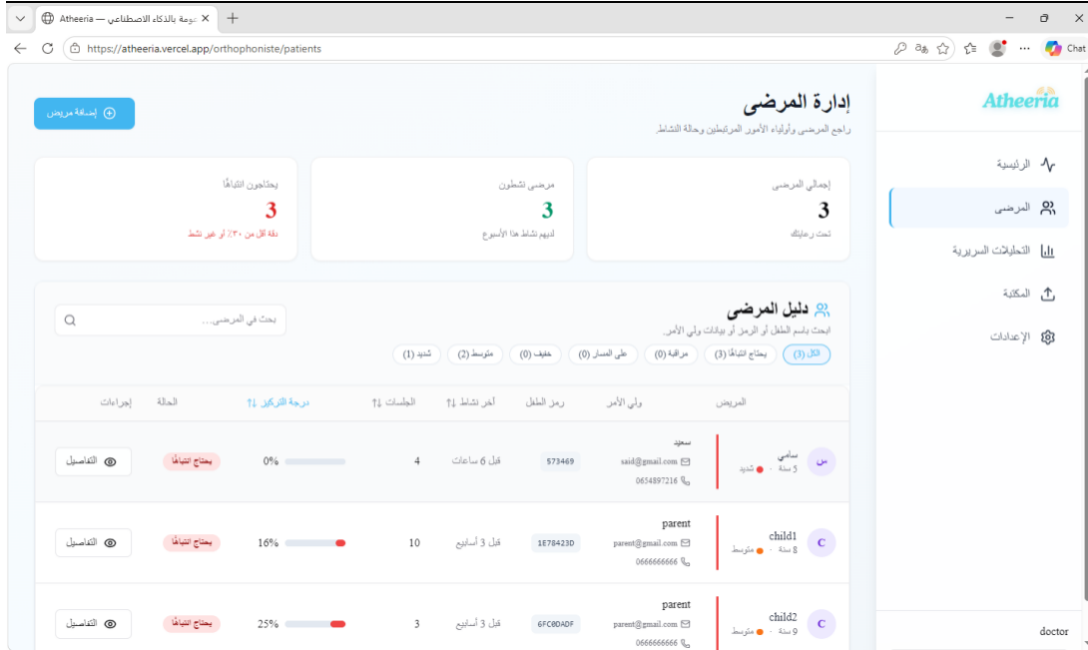


Figure 5.5: Children List

5.3.5 Intelligent Module and File Management:

Below the statistical indicators on the homepage, the application lists the educational documents (PDF files) that the parent has linked to the platform's system ("Recent Modules"). The integration of a server-side analysis algorithm not only identifies these files but also automatically calculates a "reading accuracy" for each file based on the ratio of correct answers given by the child via Alexa. Each document thus becomes a quantifiable exercise.

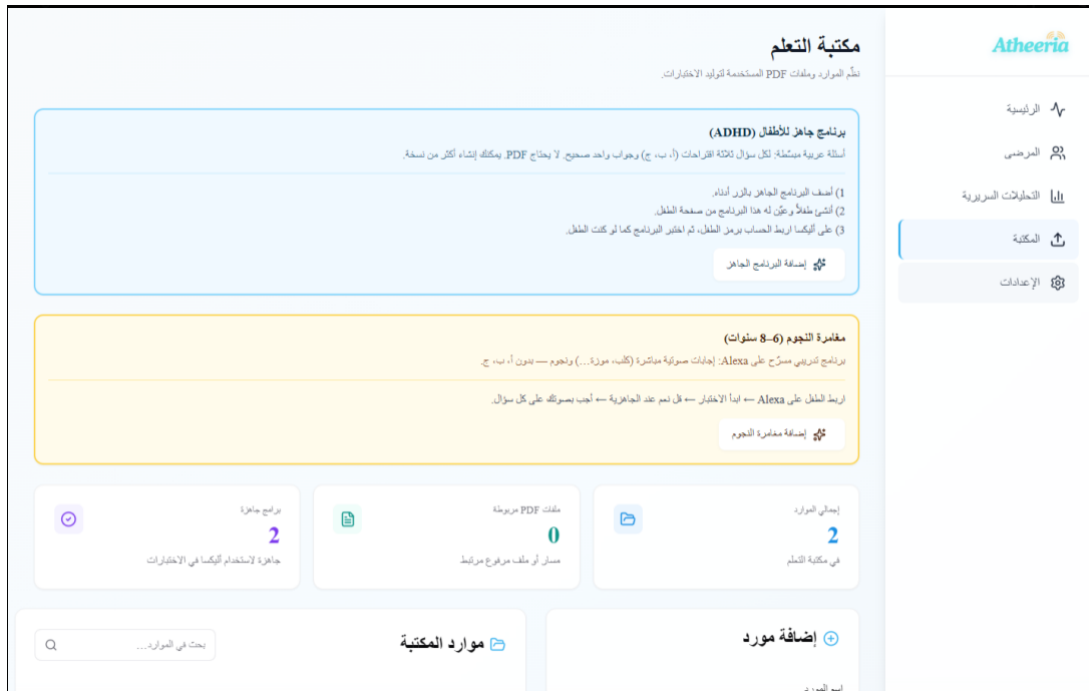


Figure 5.6: Module and File Management Interface

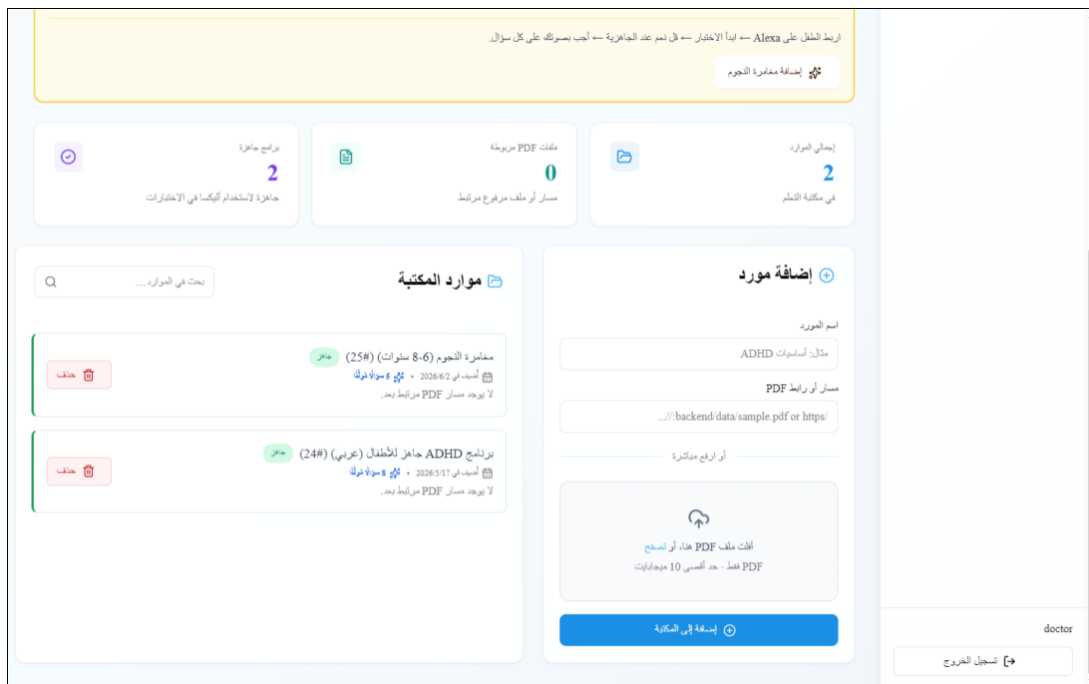


Figure 5.7: Module and File Management interface

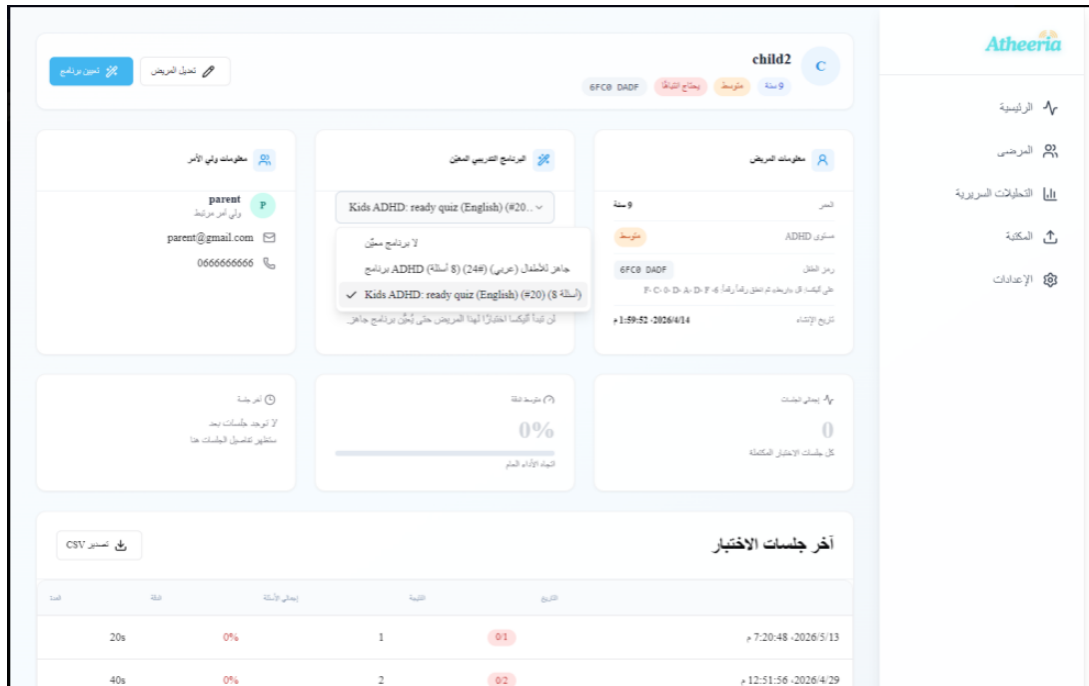


Figure 5.8: Module and File Management interface

5.3.6 Parents Dashboard

The Parents Space is a user-friendly interface designed to keep parents actively involved in their child's therapeutic journey.

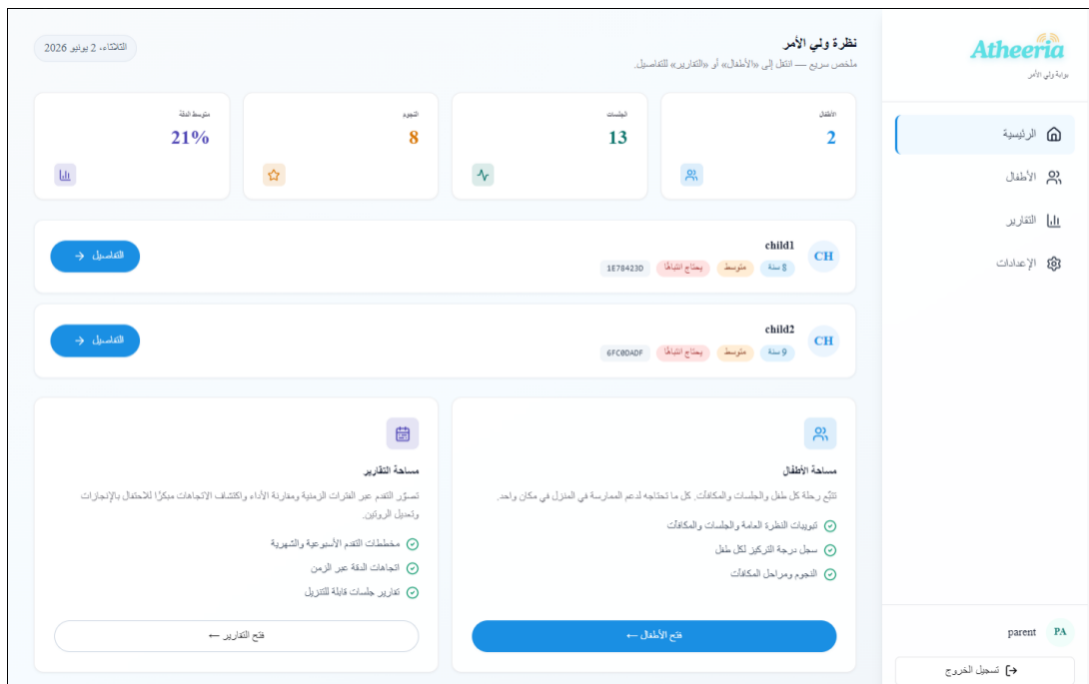


Figure 5.9: Parents Dashboard

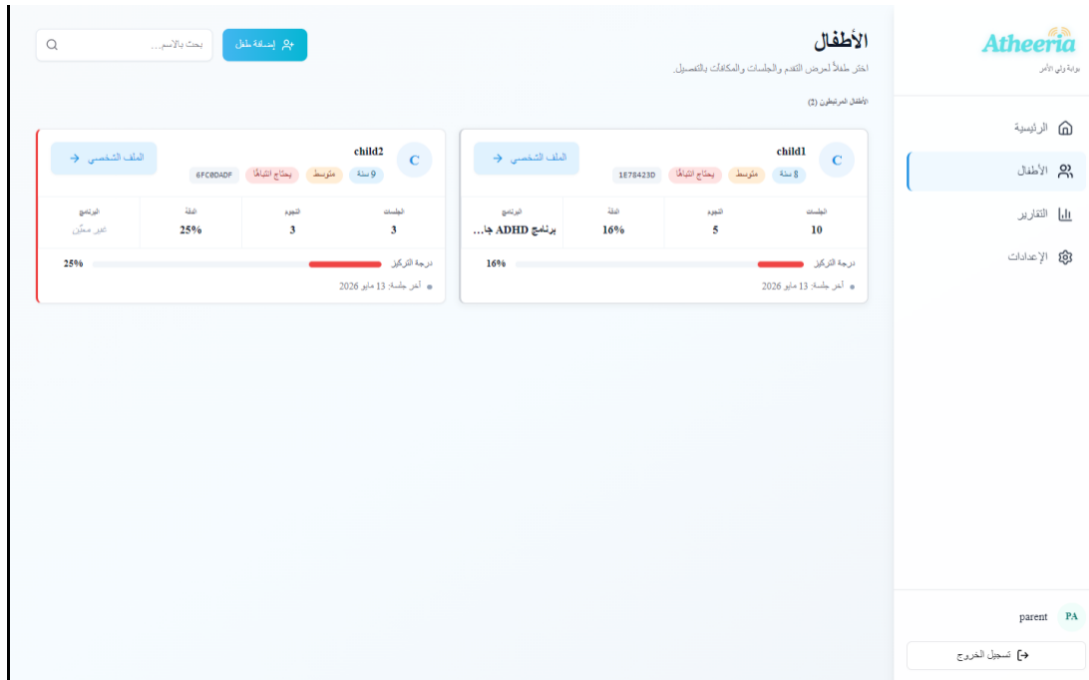


Figure 5.10: Parents Space

The screenshot shows the 'إضافة طفل' (Add Child) form in the Atheeria app. The form is titled 'أضف طفلاً جديداً ضمن حسابك الحالي' (Add a new child to your current account). It contains the following input fields:

- اسم الطفل (Child Name): Example: أم
- العمر (Age): Example: 8
- الجنس (Gender): Example: أنثى (Female)
- ملاحظة التشخيص (اختياري) (Diagnosis Note (Optional)): Example: ADHD متوسط (Moderate ADHD)

At the bottom of the form, there are two buttons: 'إضافة الطفل' (Add Child) and 'إلغاء' (Cancel). The right sidebar is identical to the one in Figure 5.10, showing navigation icons and the 'parent PA' and 'تسجيل الخروج' (Logout) buttons.

Figure 5.11: Add children



Figure 5.12: Learning statistics interface

5.3.7 The Main Dashboard

The administration using the dashboard to manage and add new users and verify subscriptions and security.

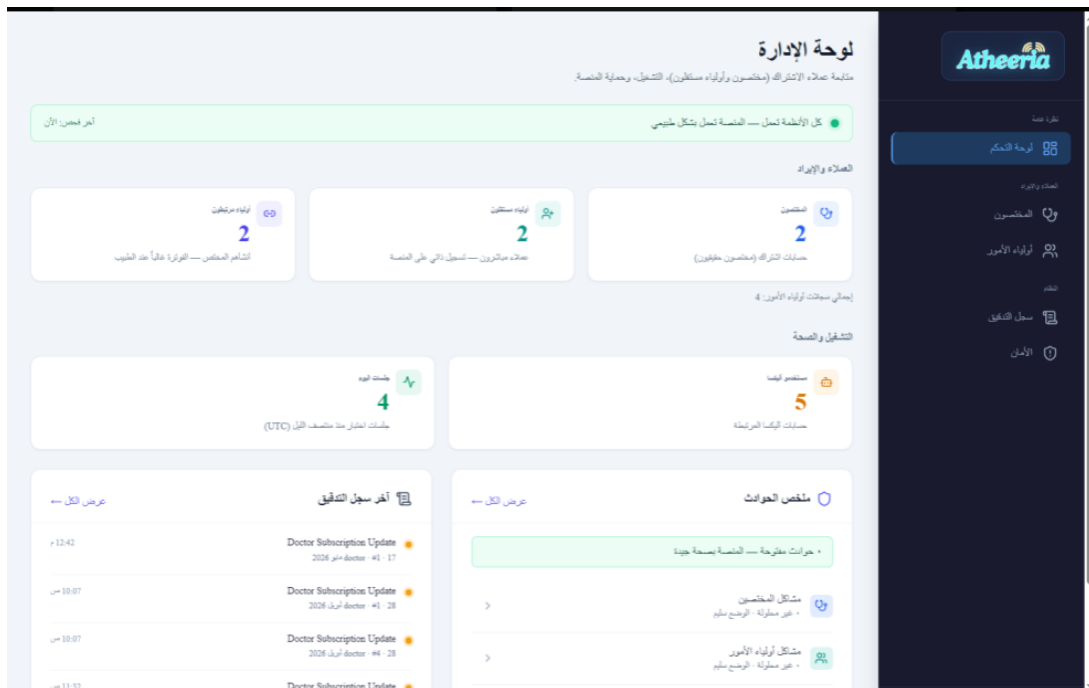


Figure 5.13: Main Admin Dashboard Page Interface (Dachboard)

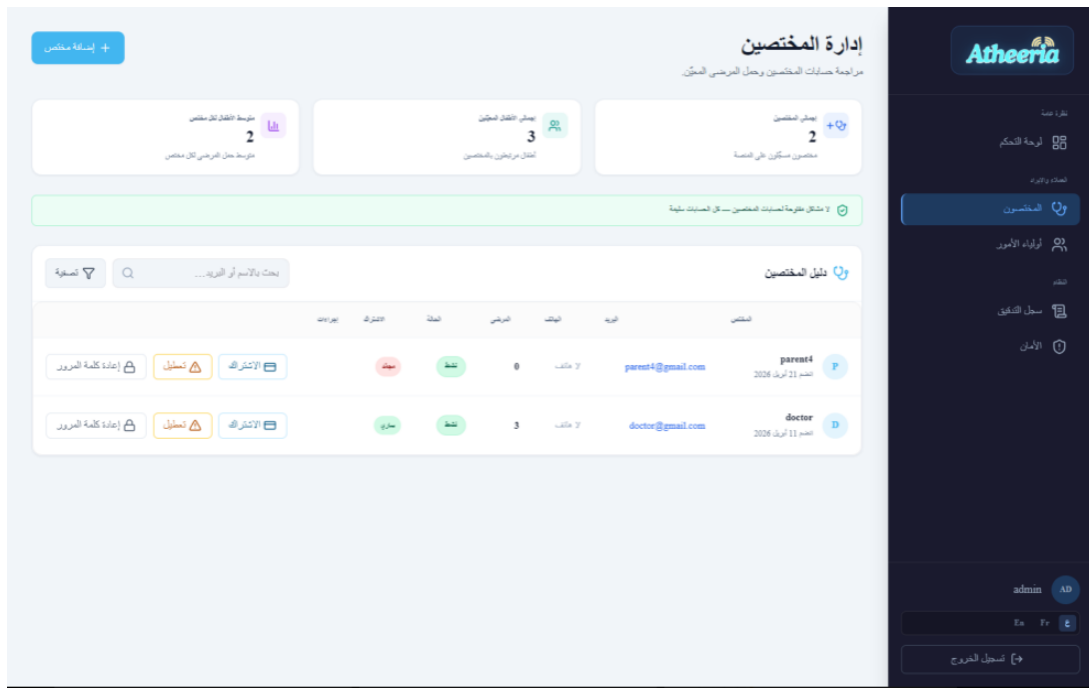


Figure 5.14: Main Interface of the Admin Dashboard page (Subscribers)

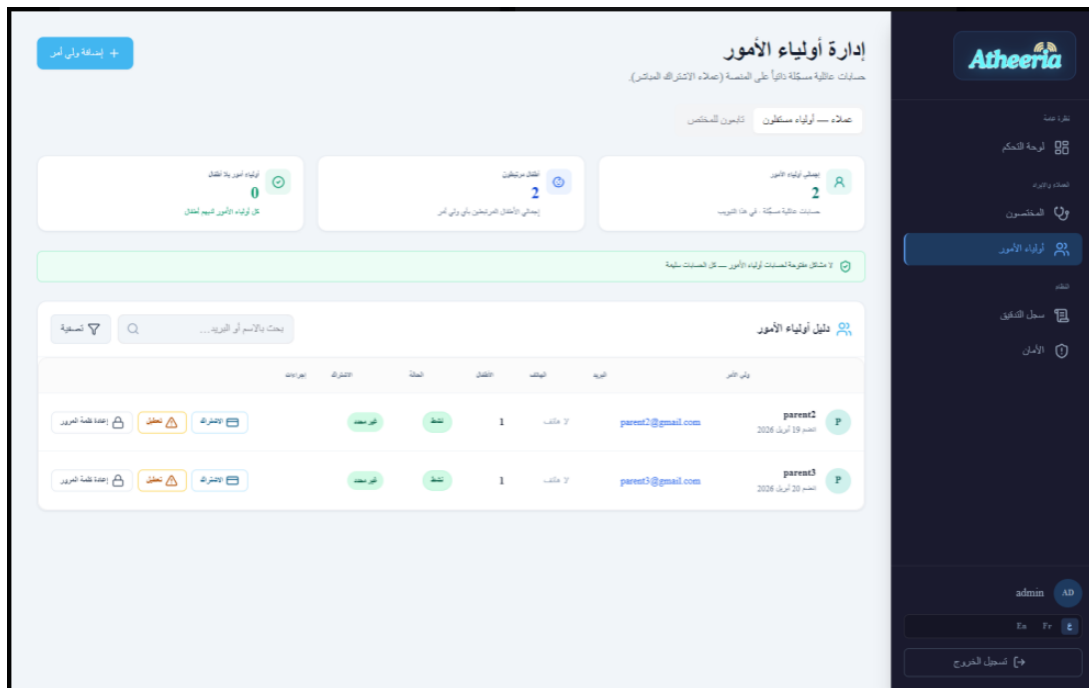


Figure 5.15: Main Interface of the Admin Dashboard page (Subscribers)

سجل التدقيق

تتبع إجراءات الإدارة وأحداث الأمان

عملية تصفية

إعادة ضبط

تطبيق

بحث في التفاصيل...

كل أنواع الأحداث

كل الإجراءات

حذف

تعطيل

تنشيط

إلغاء

تتبع

9 سجلات

الوقت	الحدث	الجهة	البيانات
12:42 - 17:05 م	doctor #1	doctor_subscription_update	..{"grace_days":8,"paid_until":"2026-06-30"}
10:07 - 28:04 ص	doctor #1	doctor_subscription_update	..{"grace_days":8,"paid_until":"2026-05-09"}
10:07 - 28:04 ص	doctor #4	doctor_subscription_update	..{"grace_days":8,"paid_until":"2026-05-09"}
11:32 - 27:04 ص	doctor #1	doctor_subscription_update	..{"grace_days":8,"paid_until":"2026-04-24"}
11:30 - 27:04 ص	doctor #4	doctor_subscription_update	..{"grace_days":8,"paid_until":"2026-04-26"}
11:30 - 27:04 ص	doctor #1	doctor_subscription_update	..{"grace_days":8,"paid_until":"2026-04-26"}

Figure 5.16: chronological record

اشتراك المختص (يد بيد)

الحالة الحالية: غير محدد

آخر يوم مشمول في الفترة المدفوعة

dd/mm/yyyy

اتركه فارغاً إن لم يُحدّد بعد. بعد هذا اليوم تبدأ مهلة السماح تم يُجمّد إقتضاء المحتوى وإسناد برامج جديدة.

أيام السماح (اختياري)

فارغ = القيمة الافتراضية من الخادم (SUBSCRIPTION_DEFAULT_GRACE_DAYS).

معفى من الفوترة على المنصة

لحسابات تجريبية أو اتفاق خاص؛ يُتجاهل تجميد الاشتراك.

حفظ

إلغاء

Figure 5.17: Subscription Verification

Chapter 6

FINANCIAL PLAN

6.1 Costs and Expenses

Nature of Expenses (DA)	N+1	N+2	N+3	N+4	N+5
electricity and gas consumption	150,000.00	200,000.00	200,000.00	200,000.00	200,000.00
Algeria Telecom service	50,000.00	100,000.00	100,000.00	150,000.00	150,000.00
rent	480,000.00	480,000.00	600,000.00	600,000.00	600,000.00
maintenance	50,000.00	50,000.00	50,000.00	50,000.00	50,000.00
insurance	300,000.00	300,000.00	400,000.00	500,000.00	500,000.00
staff expenses	6	12	17	22	29
other operating costs	865000	1410000	1847000	2353500	3041000
marketing costs	1,200,000.00	1,450,000.00	1,890,000.00	2,000,000.00	2,100,000.00
TOTAL GENERAL OF CHARGES	3095006	3990012	5087017	5853522	6641029

Figure 6.1: Summary of charges and costs

The costs and expenses of the organization are as follows:

operating costs such as renting an apartment, electricity costs and gas, Algeria Telecom, material insurance

Nature of Expenses (DA)	N+1	N+2	N+3	N+4	N+5
electricity and gas consumption	150,000.00	200,000.00	200,000.00	200,000.00	200,000.00
Algeria Telecom service	50,000.00	100,000.00	100,000.00	150,000.00	150,000.00
rent	480,000.00	480,000.00	600,000.00	600,000.00	600,000.00
maintenance	50,000.00	50,000.00	50,000.00	50,000.00	50,000.00
insurance	300,000.00	300,000.00	400,000.00	500,000.00	500,000.00
TOTAL GENERAL OF CHARGES	1030000	1130000	1350000	1500000	1500000

Figure 6.2: operating expenses

The operating costs of our solution

Cost Category (DA)	N+1	N+2	N+3	N+4	N+5
AI & LLM API Services	500,000	750,000	1,125,000	1,462,500	1,900,000
Cloud Infrastructure & Compute Servers (AWS Lambda, Hosting)	280,000	532,500	531,000	642,000	809,000
Database, Storage & Core Services (Firestore, Auth, Monitoring)	90,000	127,500	191,000	249,000	332,000
TOTAL OPERATING COSTS	870,000	1,410,000	1,847,000	2,353,500	3,041,000

Figure 6.3: The operating costs of the solution

Personnel expenses represent the details of the salaries of our staff.

Poste / Fonction	N+1			N+2			N+3			N+4			N+5		
	Salary	Effectif	Annual Total	Salary	Effectif	Annual Total	Salary	Effectif	Annual Total	Salary	Effectif	Annual Total	Salary	Effectif	Annual Total
Founder	80,000.00	2	1,920,000.00	100,000.00	2	2,400,000.00	120,000.00	2	2,880,000.00	140,000.00	2	3,360,000.00	160,000.00	2	3,840,000.00
AI / Fullstack Developer	70,000.00	2	1,680,000.00	80,000.00	4	3,840,000.00	90,000.00	4	4,320,000.00	100,000.00	6	7,200,000.00	120,000.00	6	8,640,000.00
Developer	0.00	0	0.00	60,000.00	2	1,440,000.00	70,000.00	4	3,360,000.00	80,000.00	6	5,760,000.00	90,000.00	8	8,640,000.00
Network Engineer	0.00	0	0.00	70,000.00	1	840,000.00	80,000.00	1	960,000.00	90,000.00	2	2,160,000.00	100,000.00	3	3,600,000.00
Marketing Manager	50,000.00	1	600,000.00	60,000.00	1	720,000.00	70,000.00	2	1,680,000.00	80,000.00	2	1,920,000.00	90,000.00	3	3,240,000.00
Accountant	0.00	0	0.00	40,000.00	1	480,000.00	50,000.00	1	600,000.00	60,000.00	1	720,000.00	70,000.00	1	840,000.00
Secretary	0.00	0	0.00	30,000.00	1	360,000.00	35,000.00	1	420,000.00	40,000.00	1	480,000.00	45,000.00	1	540,000.00
UX/UI Designer	80,000.00	1	960,000.00	0.00	0	0.00	57,500.00	2	1,380,000.00	65,000.00	2	1,560,000.00	65,000.00	5	3,900,000.00
Total Personnel Annual (DA)		6	5,160,000.00		12	10,080,000.00		17	15,600,000.00		22	23,160,000.00		29	33,240,000.00

Figure 6.4: Personal expenses

Tangible fixed assets:

realisation															
tangible fixed assets	N+1			n+2			n+3			n+4			n+5		
	brut	amortissement	net	brut	amortissement	net	brut	amortissement	net	brut	amortissement	net	brut	amortissement	net
Powerful processor PC	400 000,00	80 000,00	320 000,00	400 000,00	160 000,00	240 000,00	400 000,00	240 000,00	160 000,00	400 000,00	320 000,00	80 000,00	400 000,00	400 000,00	---
Standard PC	140 000,00	28 000,00	112 000,00	140 000,00	56 000,00	84 000,00	140 000,00	84 000,00	56 000,00	140 000,00	112 000,00	28 000,00	140 000,00	140 000,00	---
office	160 000,00	32 000,00	128 000,00	160 000,00	64 000,00	96 000,00	160 000,00	96 000,00	64 000,00	160 000,00	128 000,00	32 000,00	160 000,00	160 000,00	---
network equipment	40 000,00	8 000,00	32 000,00	40 000,00	160 000,00	24 000,00	4 000,00	24 000,00	160 000,00	4 000,00	32 000,00	8 000,00	4 000,00	4 000,00	---
Test Devices (Alexa)	50 000,00	10 000,00	40 000,00	50 000,00	20 000,00	30 000,00	50 000,00	30 000,00	20 000,00	50 000,00	40 000,00	10 000,00	50 000,00	50 000,00	---
miscellaneous equipment	120 000,00	24 000,00	96 000,00	120 000,00	48 000,00	72 000,00	120 000,00	72 000,00	48 000,00	120 000,00	96 000,00	24 000,00	120 000,00	120 000,00	---
total	910 000,00	182 000,00	728 000,00	910 000,00	364 000,00	546 000,00	910 000,00	546 000,00	364 000,00	910 000,00	728 000,00	182 000,00	910 000,00	910 000,00	---

Figure 6.5: Tangible fixed assets

6.2 Revenue

- Optimistic Revenue Projections

Optimistic Revenue Stream breakdown	N+1	N+2	N+3	N+4	N+5
New Hardware Sales (Units)	640	320	640	1,600	800
Hardware Revenue (6,000 DA/unit)	3,840,000	1,920,000	3,840,000	9,600,000	4,800,000
Parent Subscription months	8,000	11,200	19,200	38,400	48,000
Parent Subscription Revenue (2,000 DA/mo)	16,000,000	22,400,000	38,400,000	76,800,000	96,000,000
Orthophoniste Subscription months	2,000	2,800	4,800	9,600	12,000
Orthophoniste Subscription Revenue (4,000 DA/mo)	8,000,000	11,200,000	19,200,000	38,400,000	48,000,000
TOTAL OPTIMISTIC REVENUE	27,840,000	35,520,000	61,440,000	124,800,000	148,800,000

Figure 6.6: Optimistic Revenue

- Pessimistic Revenue Projections

Pessimistic Revenue Stream breakdown	N+1	N+2	N+3	N+4	N+5
New Hardware Sales (Units)	400	400	400	400	800
Hardware Revenue (6,000 DA/unit)	2,400,000	2,400,000	2,400,000	2,400,000	4,800,000
Parent Subscription months	4,800	9,600	14,400	19,200	28,800
Parent Subscription Revenue (2,000 DA/mo)	9,600,000	19,200,000	28,800,000	38,400,000	57,600,000
Orthophoniste Subscription months	1,200	2,400	3,600	4,800	7,200
Orthophoniste Subscription Revenue (4,000 DA/mo)	4,800,000	9,600,000	14,400,000	19,200,000	28,800,000
TOTAL PESSIMISTIC REVENUE	16,800,000	31,200,000	45,600,000	60,000,000	91,200,000

Figure 6.7: Pessimistic Revenue

- Optimistic & Pessimistic Revenue (voice assistant)

Direct Cost of Hardware Production / Purchase	N+1	N+2	N+3	N+4	N+5
Optimistic Hardware Units Bought	640	320	640	1600	800
Cost Price per unit (4,000 DA)	4,000	4,000	4,000	4,000	4,000
Total Optimistic Hardware COGS	2,560,000	1,280,000	2,560,000	6,400,000	3,200,000
Pessimistic Hardware Units Bought	400	400	400	400	800
Cost Price per unit (4,000 DA)	4,000	4,000	4,000	4,000	4,000
Total Pessimistic Hardware COGS	1,600,000	1,600,000	1,600,000	1,600,000	3,200,000

Figure 6.8: Optimistic & Pessimistic Revenue (voice assistant)

Chapter 7

BUSINESS MODEL

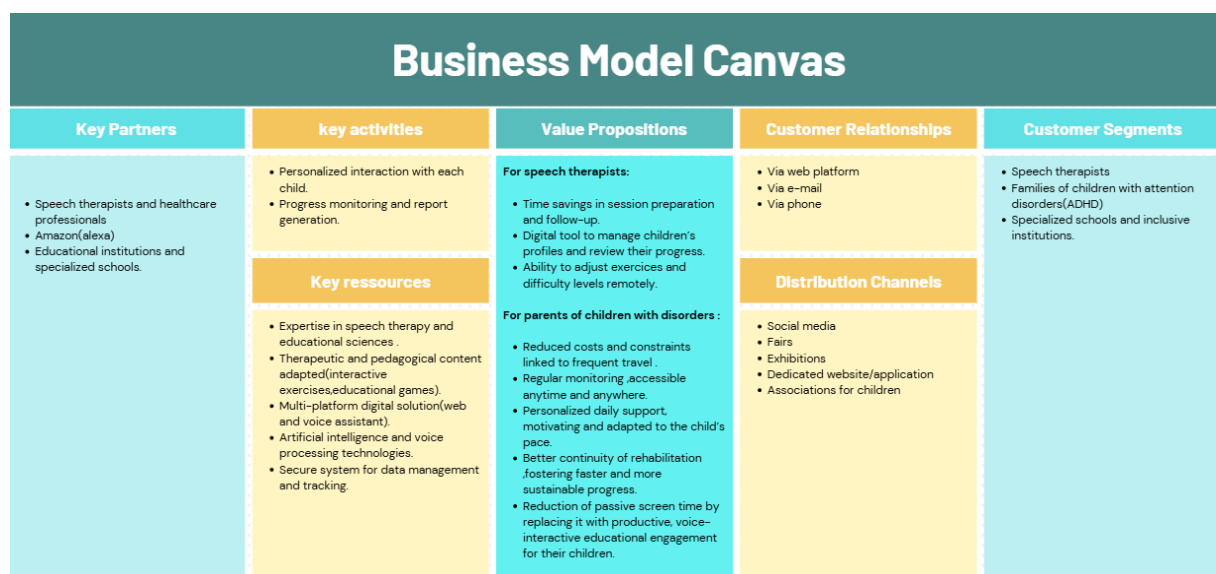


Figure 7.1: Business canvas model-part 1

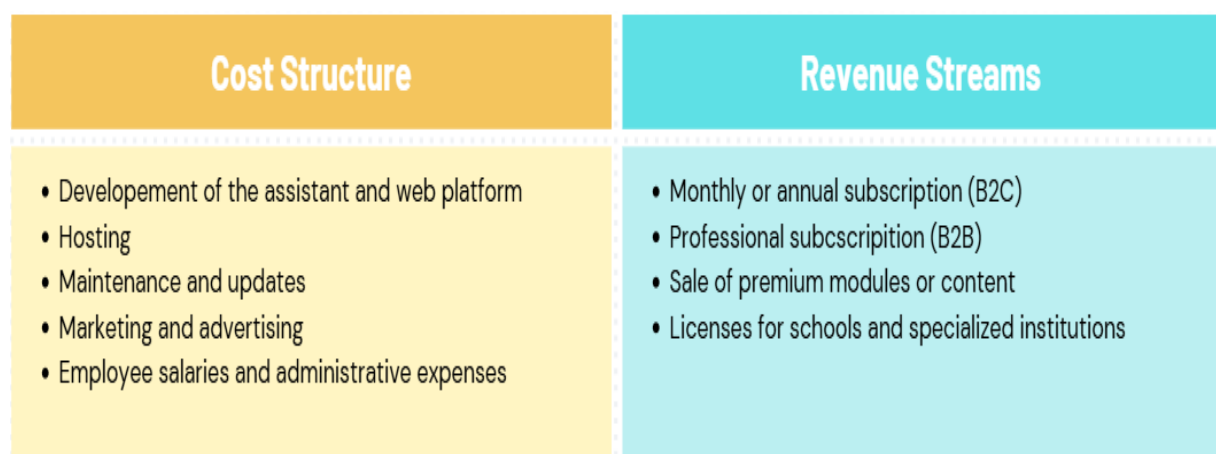


Figure 7.2: Business canvas model-part 2

CONCLUSION

In this thesis, we explored the integration of Generative AI and voice assistants to provide personalized support for children with ADHD. Through the development of our system, we aimed to transition from traditional intervention methods to intelligent, real-time behavior support systems.

Our work focused on leveraging Retrieval-Augmented Generation (RAG) to ensure that the voice assistant provides safe, clinically-grounded, and contextually relevant advice. By utilizing these technologies, we demonstrated that AI can serve as a non-intrusive companion capable of helping users manage daily routines and executive function challenges.

Furthermore, this project was developed within the framework of the university incubator, emphasizing its potential for real-world impact and innovation in the digital health sector. We analyzed the market strategy and the competitive landscape to ensure that the solution is not only technically sound but also viable as a startup project.

Future Work

Moving forward, several avenues for improvement remain:

- **Clinical Validation:** Conducting long-term user studies with healthcare professionals to measure the system’s effectiveness in therapeutic settings.
- **Enhanced Personalization:** Implementing deeper machine learning models to better predict user emotional states and adjust interventions accordingly.
- **E-payment:** Embedding secure, seamless electronic payment framework within the platform to accelerate digital adoption .

In conclusion, while the journey toward fully autonomous AI support for neurodiversity is ongoing, this project represents a significant step in utilizing modern software engineering and AI to improve the quality of life for individuals with ADHD.

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